OPEN ACCESS
Journal of Theoretical and Computational Acoustics
Vol. 29, No. 1 (2021) 2050032 (27 pages)

DOI: 10.1142/S2591728520500322

Application Limits of Conservative Source Interpolation Methods Using a Low Mach Number Hybrid Aeroacoustic Workflow

Stefan Schoder
Institute of Fundamentals and Theory in Electrical Engineering
Graz University of Technology, 8010 Graz, Austria
stefan.schoder@tugraz.at

Andreas Wurzinger, Clemens Junger and Michael Weitz
Institute of Mechanics and Mechatronics, TU Wien
1060 Vienna, Austria

Clemens Freidhager, Klaus Roppert and Manfred Kaltenbacher
Institute of Fundamentals and Theory in Electrical Engineering
Graz University of Technology, 8010 Graz, Austria

Received 5 November 2020
Accepted 16 December 2020
Published 24 February 2021

In low Mach number aeroacoustics, the well-known disparity of scales allows applying hybrid simulation models using different meshes for flow and acoustics, which leads to a fast computational procedure. The hybrid workflow of the perturbed convective wave equation involves three steps: (1) perform unsteady incompressible flow computations on a subdomain; (2) compute the acoustic sources and (3) simulate the acoustic field, using a mesh specifically suited. These aeroacoustic methods seek for a robust, conservative and computational efficient mesh-to-mesh transformation of the aeroacoustic sources. In this paper, the accuracy and the application limitations of a cell-centroid-based conservative interpolation scheme is compared to the computationally advanced cut-volume cell approach in 2D and 3D. Based on a previously validated axial fan model where spurious artifacts have been visualized, the results are evaluated systematically using a grid convergence study. To conclude, the monotonic convergence of both conservative interpolation schemes is demonstrated. Regarding arbitrary mesh deformation (for example, the motion of the vocal folds in human phonation), the study reveals that the computationally simpler cell-centroid-based conservative interpolation can be the method of choice.

Keywords: Aeroacoustics; Acoustic analogy; Hybrid CAA; Fan noise; Battery cooling fan.

1. Introduction

A key challenge in computational aeroacoustics (CAA) is the large disparity of scales between flow structures and audible acoustic wavelengths in low Mach number applications.[1] A direct numerical simulation based on the compressible flow equations, resolving both scales from the sound generating vortices to the desired evaluation position, not only leads to a high number of cells but also to a small time-step size to minimize dissipation of the acoustic waves. Hybrid schemes in CAA separate the flow from the acoustic computation using aeroacoustic analogies (see, e.g. Refs. 2 and 3). Each physical field is discretized by an optimal grid. As a result, the two grids may be quite different according to the following criteria: (1) near walls, the flow grid needs refinement to resolve boundary layers; (2) the flow grid is mostly coarsened toward the outflow boundaries to dissipate vortices and (3) the acoustic grid has to transport waves and therefore it is beneficial to use a uniform grid size all over the computational domain. The two simulations are connected by an accurate data transfer from the flow to the acoustic grid to minimize interpolation errors. Different strategies can be applied to cope with this task, starting from low complexity nearest-neighbor interpolation[4,5] to complex volume intersections between flow and acoustic grid.[6,7]

Hybrid CAA schemes, which consider just a forward coupling of the flow to the acoustic field, are valid at low Mach number if there is no feedback. For low Mach number Ma flows, the scaling between the acoustic wavelength λ and the characteristic length of a vortex l is given by $\lambda \sim l/\mathrm{Ma}$. Therefore, the sizes of the two grids are quite different. A simple nearest-neighbor interpolation for computing the acoustic sources has shortcomings (see, e.g. Refs. 4 and 5). In Refs. 4, 8 and 9, the acoustic right-hand side is computed by summing the contributions of all flow cells belonging to finite elements (FEs) surrounding an acoustic FE node. Thereby, the method assumes that the flow quantities used for the acoustic source term computation are constant over each flow cell. This approach has also been used in Ref. 10 for three-dimensional (3D) problems, where a grid dependency has been observed, resulting in an underestimated sound pressure level over the whole frequency spectrum. With respect to the constant values over each flow cell, any function-based interpolation technique to construct an enriched function space (e.g. radial basis function interpolation,[11–15] FE interpolation[16]) will improve the interpolation result and increase the computational effort.

Particularly important for the FE method (FEM) interpolation is that interpolation of the FE sources can be done in two ways. First, the field variable is interpolated to the integration points of the FE and then processed by the FE. This procedure has the drawback that it does not conserve energy.[4] Second, the whole source integral (right-hand side, RHS)

$$\mathrm{RHS}(\mathbf{x}, t) = \int_{\Omega} \psi f^{\mathrm{a}} \mathrm{d}\xi \tag{1}$$

is interpolated. Therefore, the integral properties and energy are conserved. This procedure leads to a natural conservative interpolation having the same computational burden as an ordinary field interpolation. Therefore, the conservative approaches presented follow the second idea that the integrated properties are conserved; however, we assume that the lowest-order function space (space of constant functions[a]) for the flow field is sufficient, which leads to a fast and efficient algorithm.

Regarding the conservation idea, a globally conservative approach (cell-centroid-based approach) has been developed in Ref. 16, where the acoustic sources within the FE formulation are first computed on the fine flow grid. These so-called loads are then interpolated to the acoustic grid with a nearest-neighbor interpolation. This approach works with high accuracy in cases where the flow grid has smaller cells than the acoustic grid. However, in cases where the flow grid gets coarser than the acoustic grid, an approach like this induces a phase error into the interpolated field. Then, the sources look like a chequerboard (Fig. 1 on the right-hand side). Regions, where the flow grid is coarser than the acoustic mesh, occur mostly for the coarsened flow grid toward inflow and outflow boundaries since the acoustic grid has to transport waves and therefore needs a uniform grid size all over the computational domain.

To overcome such problems in the context of FEs, we have developed the cut-volume cell weight interpolation and applied it to the aeroacoustic computation of an axial fan.[17] Regarding finite volume (FV) schemes for acoustic propagation, similar investigations were performed in Ref. 5. To accelerate the calculation for volume sources within a boundary element (BE) method solving Lighthill's inhomogeneous wave equation, the authors in Ref. 18 spatially condense the acoustic source data.

The first application of the cut-volume cell weight interpolation was performed in Ref. 17. Building upon the idea, we study the acoustic effects of the current approach compared to the simpler cell-centroid-based approach.[16] Therefore, the algorithms are outlined step-by-step using pseudocode. We aim to prove the following four hypotheses: First the cell-centroid-based approach works with high accuracy in cases where the flow grid has smaller cells than the acoustic grid. Second, we aim to detect a lower bound of the acoustic element size concerning the flow cell for both methods. Third and for computational efficiency,

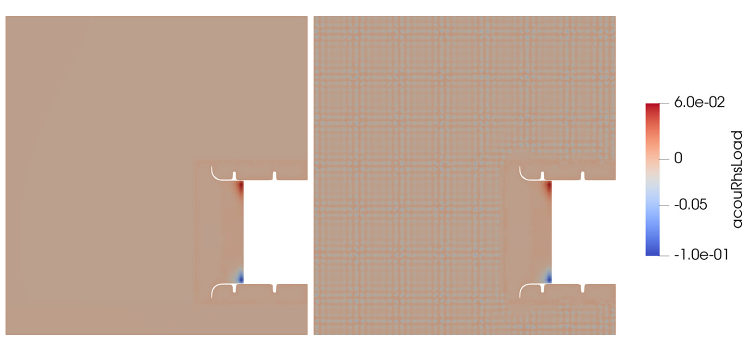


Fig. 1. Comparison of the nodal values using the cell-centroid (right) based and the cut-volume cell (left) interpolation algorithm in areas where the flow grid is coarser than the acoustic mesh.

[a]If a higher-order representation is desired, an additional regression step must be executed to construct the higher-order representation,[6] which reduces the computational efficiency.

we would like to detect the maximum potential coarsening of the acoustic element size concerning the flow field. Fourth, we study the influence of an awkward chequerboard visualization on the sound. Having investigated these hypotheses, we can provide a general guideline for choosing the most suitable method for your hybrid aeroacoustic workflow. This study clarifies when an application of the cut-volume cell weight interpolation is more beneficial, considering the computational performance and the accuracy of the interpolation.

This paper is organized as follows: In Secs. 2 and 3, we provide the aeroacoustic model, describe the two interpolation approaches and their limitations considering a variation of the CAA grid. In Secs. 4 and 5, we study the effects of the two different conservative interpolation schemes on the acoustic results concerning the acoustic discretization. Upon previous findings, we exclusively concentrate on the interpolation between the flow and the acoustic mesh through a detailed mesh study. We show the increase in computational performance through coarsening the acoustic mesh without losing numerical accuracy and discuss the suitability of the applied interpolation methods. Finally, in Sec. 6, we conclude the findings and give additional remarks for future research.

2. Formulation

Having the definitions of acoustic perturbation equations (APEs)[19–23] and introducing an arbitrary Lagrangian–Eulerian (ALE) description for the operators $\frac{D}{Dt} = \frac{\partial}{\partial t} + (\overline{\mathbf{v}} - \mathbf{v}_{\mathrm{r}}) \cdot \nabla$, where $\mathbf{v}_{\mathrm{r}}$ is the relative velocity of the grid, we arrive at the perturbed convective wave equation (PCWE) for rotating systems (see Ref. 17)

$$\frac{1}{c^2}\frac{D^2\psi^{\mathrm{a}}}{Dt^2} - \Delta\psi^{\mathrm{a}} = -\frac{1}{\bar{\rho}c^2}\frac{Dp^{\mathrm{ic}}}{Dt}. \tag{2}$$

This scalar convective wave equation is computationally efficient and describes aeroacoustic sources generated by incompressible flow structures and wave propagation through moving media. This formulation reduces the number of unknowns (acoustic pressure $p^{\mathrm{a}} = \bar{\rho}\frac{D\psi^{\mathrm{a}}}{Dt}$ and particle velocity $\mathbf{v}^{\mathrm{a}} = -\nabla\psi^{\mathrm{a}}$) to just a scalar unknown, the acoustic velocity potential ψ^{a}. It should be noted that (2) is an exact reformulation of the APEs for low Mach numbers.[21] Based on the PCWE, we verify the source term interpolation of aeroacoustic sources for nonmoving and moving meshes.

The derivation of the weak formulation (1) for the FEM involves a multiplication with a test function $\psi \in H^1(\Omega)$ and an integration over the computational domain Ω. Here, the aeroacoustic source $f^{\mathrm{a}} := -\frac{1}{\bar{\rho}c^2}\frac{Dp^{\mathrm{ic}}}{Dt}$ enters the formulation. To compute the aeroacoustic sources on the acoustic grid in a hybrid workflow, (1) is approximated on the flow grid (superscript f) and then interpolated to the acoustic grid (superscript a). Naturally, this leads to an interpolation that conserves the integral property of the aeroacoustic sources used by the FE simulation.

2.1. *Cell-centroid-based source term interpolation*

Starting from (1), the cell-centroid-based approach conserves the energy globally but approximates the local energy conservation of the FE right-hand side by

$$F_i^{\mathrm{a}} = \int_{E^{\mathrm{a}}} N_i^{\mathrm{a}}(\boldsymbol{\xi}) f^{\mathrm{a}} \mathrm{d}\xi \approx \sum_{k \in M^{\mathrm{f}}} N_i^{\mathrm{a}}(\boldsymbol{\xi}_{E_k^{\mathrm{f}}}) F_k^{\mathrm{f}}, \tag{3}$$

where E_k^{f} denotes the cell of the flow grid, $\boldsymbol{\xi}_{E_k^{\mathrm{f}}}$ the local coordinate, M^{f} the number of flow cells, N^{a} the FE basis function on acoustic grid and the subscript i the node number on the CAA grid. To preserve the acoustic energy, the loads F_k^{f} of the fine flow grid are interpolated to the coarser acoustic grid (see Fig. 2(a)).

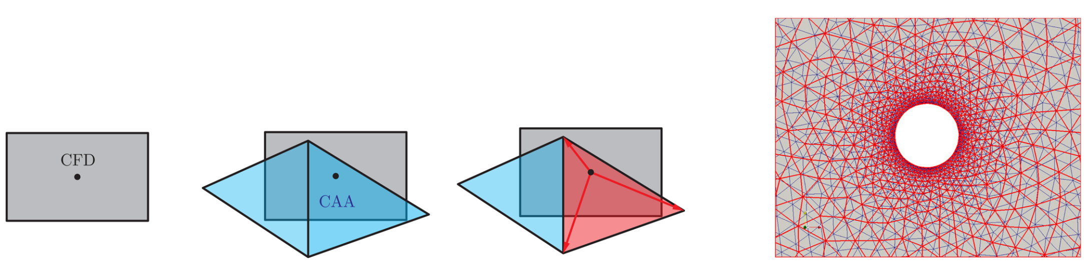
For an acoustic element, the cell-centroid-based approach (Algorithm 1) includes all loads F_k^{f} that are inside the acoustic FE E^{a}. This geometrical connection from load location to the acoustic element is accounted for by the set M^{f}. Based on the geometrical properties, the loads are weighted by the FE basis functions and summed up onto the acoustic grid. The loads

$$F_k^{\mathrm{f}} = \bigwedge_{l \to k} \int_{E_l^{\mathrm{f}}} N_l^{\mathrm{f}}(\boldsymbol{\xi}) f^{\mathrm{f}}(\boldsymbol{\xi}) \mathrm{d}\xi \tag{4}$$

are assembled on the CFD mesh in terms of the FEM, by the assembly operator $\bigwedge$. To reduce the computational complexity of (4) and to construct a computationally efficient algorithm, we simplify the integration over the source volume $V_{E_k^{\mathrm{f}}}$ to a first-order volume weighting

$$F_k^{\mathrm{f}} = V_{E_k^{\mathrm{f}}} f_k^{\mathrm{f}}(\boldsymbol{\xi}_{E_k^{\mathrm{f}}}) \tag{5}$$

and avoid the FE assembly operator. The error introduced by constant flow cell quantities may be estimated based on a Taylor series and is bounded by the product of the characteristic grid size and the gradient of the interpolated quantity, which is assumed to be small for a high-quality flow grid. As illustrated in Fig. 2(a), the algorithm has to find the nodal source



(a) Cell-centroid-based approach for conservative interpolation. Based on the cell-centroid of the flow cell, an acoustic element is searched and used to assemble the computed sources.

(b) CFD grid (red) and CAA grid (blue).

Fig. 2. (Color online) Cell-centroid-based approach for conservative interpolation and general mesh sizes of flow and acoustic grid.

location $\mathbf{x}_k$ inside an acoustic element (set M^{f}). This global position $\mathbf{x}_k$ corresponds to a local position $\boldsymbol{\xi}_{E^{\mathrm{f}}_k}$ in the reference FE ($\mathbf{x}_k \multimap \boldsymbol{\xi}_{E^{\mathrm{f}}_k}$). Finally, the loads F^{f}_k are interpolated to the nodes of the acoustic mesh using the FE basis functions N^{a}_i.

Considering Fig. 2(a), only one of the two CAA elements will receive values during the interpolation. In this example, the right element that contains the center location of the flow cell. Consequently, there could be elements having zero nodal values. To describe the observed redistribution of sources, we define a frequency–wave number relation (FWR) for a physical quantity (e.g. acoustic pressure, aeroacoustic source term)

$$\omega = \omega(\|\mathbf{k}\|_2), \tag{6}$$

where ω is the frequency, $\mathbf{k}$ the wave number and $\|\cdot\|_2$ the Euclidean norm; this relation is similar to the dispersion relation for hyperbolic differential operators. However, the limited capability of this interpolation scheme destroys the FWR of the interpolated aeroacoustic sources $\omega \neq \omega(\|\mathbf{k}\|_2)$. We conclude, if target elements receive zero values, then we expect deviations in the underlying wave number of the interpolated field. As a consequence, we see chequerboard oscillation inside the aeroacoustic sources (see Fig. 1).

This cell-centroid-based approach works accurately in cases where the flow grid is finer than the acoustic grid.[16] However, the mesh size of flow computations varies from fine meshes resolving boundary layers to coarse meshes toward regions without flow gradients. Consequently, hybrid aeroacoustics deals with regions where the size of the flow grid is equal to or larger than the acoustic mesh (see Fig. 2(b)). In these cases, the cell-centroid-based approach fails locally because it neglects the contributions weighted by the volume.[17]

2.2. *Cut-volume cell-based source term interpolation*

Thus, the improved approach considers for each acoustic volume element E^{a}_l the set of flow cells E^{f} that intersect

$$E^{\mathrm{f}\cap\mathrm{a}}_l = E^{\mathrm{a}}_l \cap E^{\mathrm{f}} \tag{7}$$

Algorithm 1. Cell-centroid-based interpolation

```
1: F^a ← CentroidInterpolation(E^a)

2: function CentroidInterpolation
3:   for i in E^a do
4:     M^f = GetAllCFDCellsOfCAAElementByCFDCellCenter(i)   ▷ M^f holds indices of CFD Cells
5:     for k in M^f do
6:       V_{E^f_k} = GetVolumeOfCFDCell(k)
7:       F^f_k = ComputeFEMLoadOnCFD(k)   ▷ Compute (5)
8:       x_k = GetCentroidOfCFDCell(k)
9:       F^a_i = InterpolateFEMLoadToCAAMesh(x_k)   ▷ Finite element interpolation to CAA element (3)
10:    end for
11:  end for
12: end function
```

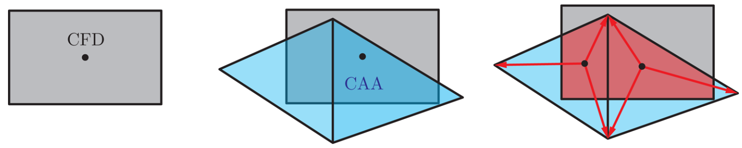


Fig. 3. Steps for intersection and interpolation based on cut-volume cell approach. Based on the flow cell, the acoustic elements are intersected. Then, the intersected elements are used to assemble the computed sources.

with the respective acoustic volume element.[6] This improved approach conserves the energy globally as well as locally for different mesh sizes and calculates the FE right-hand side by

$$\int_{E^{\mathrm{a}}} N_i^{\mathrm{a}}(\boldsymbol{\xi}) f^{\mathrm{a}} \mathrm{d}\xi = \sum_{k \in M^{\mathrm{f}\cap\mathrm{a}}} N_i^{\mathrm{a}}(\boldsymbol{\xi}_k) F_k^{\mathrm{f}\cap\mathrm{a}}. \tag{8}$$

Based on this intersection, the loads $F_k^{\mathrm{f}\cap\mathrm{a}}$ are integrated over the intersection volume $V_{\mathrm{c}} \multimap V_{E_l^{\mathrm{f}\cap\mathrm{a}}}$,

$$F_k^{\mathrm{f}\cap\mathrm{a}} = \bigwedge_{l \to k} \int_{E_l^{\mathrm{f}\cap\mathrm{a}}} N_l^{\mathrm{f}}(\boldsymbol{\xi}) f^{\mathrm{f}}(\boldsymbol{\xi}) \mathrm{d}\xi. \tag{9}$$

To avoid the FE assembly operator in (9), we assume that the aeroacoustic source term is constant over the fluid cell. Therefore, the integral reduces to a multiplication of the intersection volume $V_{E_l^{\mathrm{f}\cap\mathrm{a}}}$ with the source density f_l^{f} at the volumetric centroid $\boldsymbol{\xi}_{E_l^{\mathrm{f}\cap\mathrm{a}}}$ of the intersection polyhedron

$$F_k^{\mathrm{f}\cap\mathrm{a}} = V_{E_k^{\mathrm{f}\cap\mathrm{a}}} f_k^{\mathrm{f}}(\boldsymbol{\xi}_{E_k^{\mathrm{f}\cap\mathrm{a}}}). \tag{10}$$

The algorithmic workflow is depicted in Fig. 3. As opposed to the cell-centroid-based approach, we determine two intersecting cells; we compute the volumetric center $\boldsymbol{\xi}_{E_l^{\mathrm{f}\cap\mathrm{a}}}$ and the volume of the intersection polyhedron $V_{E_l^{\mathrm{f}\cap\mathrm{a}}}$. This additional intersection information allows us to add the source contribution weighted with the intersection volume to the nodes of the acoustic element using the FE basis functions evaluated at the center of the polyhedron (see Fig. 3). Thereby, we cover cases in which acoustic elements are smaller and embedded into a single flow cell. We conclude that all target elements receive source values and we do not see chequerboard oscillation inside the aeroacoustic sources (see Fig. 1).

3. Discretization Considerations

In this section, we study the first three hypotheses given in Sec. 1. We will therefore start with the accuracy of the source interpolation for variable mesh combinations and obtain a lower bound for the acoustic mesh.

Algorithm 2. Cut-volume cell interpolation

$\mathbf{F}^{\mathrm{a}} \leftarrow \mathrm{CutVolumeCellInterpolation}(E^{\mathrm{a}})$

function CutVolumeCellInterpolation
  for i in E^{a} **do**
    $M^{\mathrm{f}\cap\mathrm{a}} = \mathrm{IntersectCFDGridWithCAACell}(i)$ ▷ $M^{\mathrm{f}\cap\mathrm{a}}$ holds indices of CFD Cells
    for k in $M^{\mathrm{f}\cap\mathrm{a}}$ **do**
      $V_{E_l^{\mathrm{f}\cap\mathrm{a}}} = \mathrm{GetIntersectionVolumeOfCFDCell(k)}$
      $F_k^{\mathrm{f}\cap\mathrm{a}} = \mathrm{ComputeFEMLoadOnCFD(k)}$ ▷ Compute (10)
      $\boldsymbol{x}_k = \mathrm{GetCentroidOfIntersectionCell(k)}$
      $F_i^{\mathrm{a}} = \mathrm{InterpolateFEMLoadToCAAMesh}(\boldsymbol{x}_k)$ ▷ Finite element interpolation to CAA element (8)
    end for
  end for
end function

3.1. *Limitation for fine CAA meshes — visualization*

The cell-centroid-based conservative interpolation and the cut-volume cell approach are compared and verified against an analytic source function $f(\mathbf{x}) = \sin(3\pi x)$ on the domain $\Omega(x, y) \in [0, 1]^2$. Considering the cell-centroid-based approach, errors will occur for large mesh ratios Γ

$$\Gamma = \frac{N_{\mathrm{CAA}}}{N_{\mathrm{CFD}}}, \tag{11}$$

where N_{CAA} is the number of nodes along the CAA mesh and N_{CFD} is the number of nodes along the CFD mesh. Figure 4 shows how the FE nodal right-hand side values develop on a line at $y = 0.5$. Along the evaluation line, the number of nodes of the CFD grid is constant for the different evaluations (101 nodes). In contrast to that, the number of nodes along the CAA mesh varies according to the mesh ratio. If $\Gamma < 1$, the CFD grid is finer than the CAA mesh[b]; for $\Gamma > 1$, the CFD grid is coarser than the CAA mesh.[c] The cell-centroid-based interpolation performs satisfactorily for low mesh ratios where the CFD grid is finer than the CAA mesh. For larger mesh ratios, the cell-centroid-based interpolation exhibits spurious modes and energy is transferred to shorter wavelengths (unphysical). Overall, the cut-volume cell approach has the desired conservative properties for all mesh ratios.

Based on Parceval's theorem, the relative energy content of the real wavelength $\lambda_{\mathrm{a}} = 2/3\,\mathrm{m}$ compared to the total energy is evaluated by the energy ratio e_{r}

$$e_{\mathrm{r}} = \frac{(\hat{D}_{\mathrm{RHS}}(\lambda_{\mathrm{a}}))^2}{\sum_{k=1}^{N}(\hat{D}_{\mathrm{RHS}}(\lambda_k))^2}. \tag{12}$$

[b]This is the standard case when doing aeroacoustic simulations.
[c]This event can occur toward inflow and outflow boundaries for incompressible flow simulations and using a hybrid aeroacoustic workflow.

Table 1. Ratio of the energy corresponding to the actual wavelength and the total energy for the cell-centroid -based procedure and the cut-volume cell procedure, respectively.

Γ	Coverage ratio ε	Cell-centroid e_{r}	Cut-volume cell e_{r}	CPU time cell-centroid	CPU time cut-volume cell
1/30	99.99%	99.31%	99.27%	0.278 s	3.26 s
0.5	99.98%	99.28%	99.22%	0.282 s	4.39 s
1	87.18%	99.17%	98.92%	0.279 s	6.61 s
2	50.29%	85.48%	98.79%	0.368 s	10.983 s
5	20.2%	52.55%	98.85%	0.562 s	23.938 s
10	10.1%	31.72%	98.94%	0.835 s	46.158 s
20	5.05%	15.73%	99.04%	1.457 s	88.403 s
30	3.37%	11.03%	99.1%	2.048 s	129.853 s

$\hat{D}_{\mathrm{RHS}}$ are the spatial amplitudes of the Fourier transformed right-hand side of (1) along the spatial coordinate x. Table 1 compares the cell-centroid-based conservative interpolation (cell-centroid) to the conservative cut-volume cell interpolation (cut-volume cell). As expected, the more accurate conservative cut-volume cell interpolation significantly outperforms the cell-centroid-based procedure when it comes to a source visualization. However, the computational demand increases with the number of cell intersections. Compared to the setup time of the cell-centroid-based approach, the setup time for the cut-volume cell intersection takes about 50 times as long but overall, the computational demand increases linearly.

Additionally to the mesh ratio Γ, Table 1 lists the coverage ratio ε, which describes the ratio of CAA cells that are covered by CFD cells relative to the total number of CAA cells. For example, the CAA mesh in Fig. 2(a) has a coverage ratio of $\varepsilon = 0.5$; reddish cells are covered and bluish cells are not covered. The energy ratio e_{r} for the cell-centroid-based interpolation decreases with decreasing coverage ratio ε. Closer examination of the relation between e_{r} for the cell-centroid-based interpolation and ε revealed that $e_{\mathrm{r}} \sim 1 - (1 - \varepsilon)^3$. Regarding this first analysis, we found that the first hypothesis holds. Indeed, the cell-centroid-based approach works with high accuracy in cases where the flow grid has smaller cells than the acoustic grid. But we still have to outline the consequences of this source redistribution inside the acoustic spectrum.

For large mesh ratios Γ, the cell-centroid interpolation fails to conserve the fluid dynamic sources locally. In this case, the CAA mesh re-samples the contributions of the CFD grid and redistributes them toward wrong wave numbers $\mathbf{k}$ that are connected to the spatial occurrence of the CFD values and the CFD mesh. The mesh dependency was already reported.[10] These high wave number oscillations increase in size and extend for increasing mesh ratios Γ (see Fig. 4).

3.2. *Limitation for fine CAA meshes — propagation*

However, the error contribution inside the final propagation simulation is unclear yet. Thus, we try to explain the redistribution effect on sound propagation. When propagating these

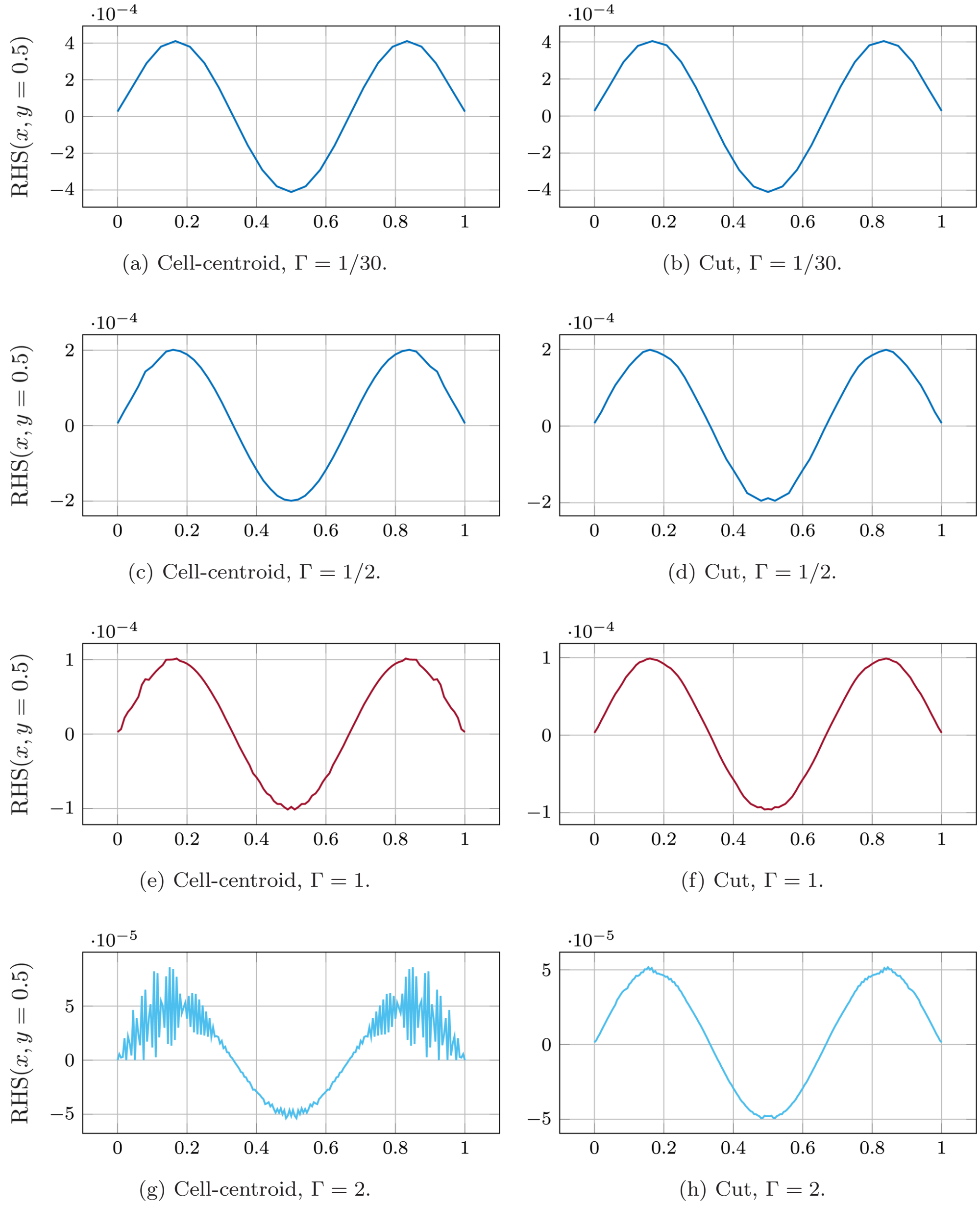
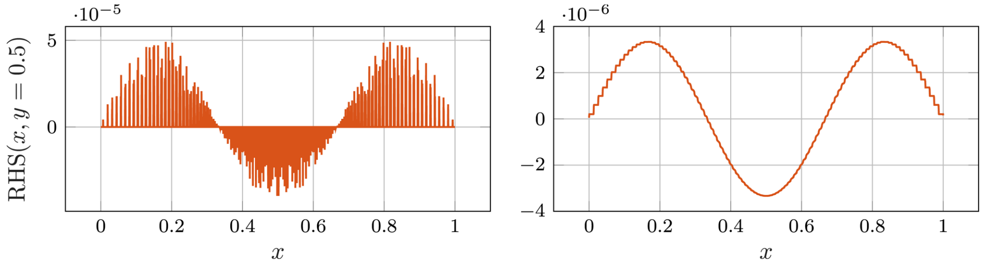


Fig. 4. Right-hand side values (1) at $y = 0.5$ using the flow source density $f(\mathbf{x}) = \sin(3\pi x)$. The number of nodes of the CFD grid along the evaluation line is 101; whereas the number of nodes of the CAA grid along the line is based on the mesh ratio Γ. $\Gamma < 1$: the CFD grid is finer than the CAA mesh; $\Gamma > 1$: the CFD grid is coarser than the CAA mesh. The cell-centroid-based interpolation for different mesh ratios is illustrated in (a, c, e, g and i) and the cut-volume cell approach is illustrated in (b, d, f, h and j).



(i) Cell-centroid, $\Gamma = 30$.

(j) Cut, $\Gamma = 30$.

Fig. 4. (*Continued*)

redistributed aeroacoustic sources, the distribution will be shifted onto the dispersion relation (according to the wave operator) for the specific frequency, with a certain amount of phase error $\Delta\varphi_{\text{error}}$. Since the wave operator enforces the dispersion relation of the particular wave model. The redistributed source peaks propagate waves that are then superposed. Therefore, we subdivide the first hypothesis into two subhypotheses. Hypothesis 1a: For low source shifts $\Delta s < \lambda$, the impact of this redistribution on the wave propagation is negligible. Hypothesis 1b: For larger source shifts $\Delta s \approx \lambda$, where the acoustic wavelength and the characteristic length of the flow converge, the impact on the phase error increases, and constructive or destructive interference can occur.

Regarding the cell-centroid-based approach, the source shift Δs_{cc} for the corresponding elements yields

$$\Delta s_{\text{cc}} = \max\left(\min \frac{c_{\text{CAA}} - c_{\text{CFD}}}{\lambda}, \frac{|\Delta x_{\text{CAA}} - \Delta x_{\text{CFD}}|}{\lambda}\right), \tag{13}$$

with c_{CAA} being the center location of the CAA cell, c_{CFD} being the center location of the CFD cell, Δx_{CAA} the characteristic cell size of an CAA element and Δx_{CFD} the characteristic cell size of an CFD element. The maximum shift Δs_{cc} depends on the different location of the cells since no intersection is computed. Furthermore, it also depends in the absolute maximum size difference between the CFD and the CAA cell because smaller CAA cells then CFD cells cause a redistribution since no intersection is computed.

Regarding the cut-volume cell-based approach, the source shift Δs_{cvc} for intersecting elements yields

$$\Delta s_{\text{cvc}} = \max\left(0, \frac{\Delta x_{\text{CAA}} - \Delta x_{\text{CFD}}}{\lambda}\right). \tag{14}$$

The maximum shift Δs_{cvc} depends only on the maximum difference of the characteristic cell size if the CAA cells are larger. For any other case, no redistribution is present. To illustrate the two hypotheses, we can bound the phase error $\Delta\varphi_{\text{error}}$

$$\Delta\varphi_{\text{error}} \leq c_1 \Delta s. \tag{15}$$

Additionally, we can identify the source shift as the Helmholtz number of the relocation $\text{He}_{\text{r}} = \Delta s$.

Hypothesis 1a. Each source excites a pressure wave that is then superposed to the acoustic pressure p^{a}. If the source shift is acoustically compact, with Helmholtz number $\mathrm{He_r} = \Delta s \ll 1$, we conclude an insignificant difference of the resulting acoustic field, regarding the interpolation algorithm. Then, the phase error $\Delta\varphi_{\mathrm{error}}$ of the redistributed sources is negligible.

Hypothesis 1b. However, if the source redistribution violates compactness $\mathrm{He_r} = \Delta s \sim 1$, the phase error is significant. Then, as proposed in Hypothesis 1b, the wrongly distributed sources modulate the sound oddly, and constructive or destructive interference can occur. Additionally, mitigating this phase error close to aeroacoustic sources can be important for the acoustic near-field. To conclude, phase errors are more likely for the cell-centroid-based interpolation.

Hypothesis 2. According to Hypotheses 1a and 1b, the lower bound of the cell-centroid-based interpolation approach is a combination of the mesh and the resolved acoustic quantity $\mathrm{He_r} \ll 1$. But to be on the safe side, if you detect a chequerboard source distribution use the cut-volume cell approach.

Hypothesis 3. According to Hypotheses 1a and 1b, we found an explanation when the influence of an awkward chequerboard visualization on the acoustic results is occurring concerning the grid compactness, measured by $\mathrm{He_r}$.

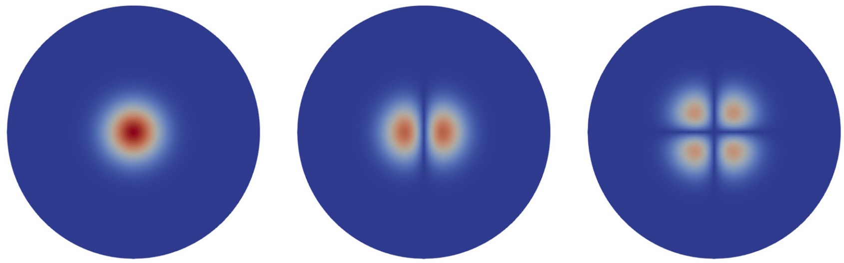
3.3. *Limitation for coarse CAA meshes*

In this section, we investigate the possible coarseness of the acoustic grid, compared to the CFD grid. Therefore, the cut-volume cell and the cell-centroid approach are verified numerically for generic source distributions (see Table 2). These distributions correspond to a monopole, dipole and quadrupole and are determined solely by the coordinates x and y, and the spatial parameter $\sigma = 0.75\,\mathrm{m}$. For a source exhibiting monopole radiation f_{MP}, the spatial parameter σ is in the range of the displacement thickness. Considering the dipole sources f_{DP}, the spatial parameter σ is the distribution of a pressure loading of a surface and in the case of a quadrupole f_{QP}, the spatial parameter σ is a measure for the eddy diameter.

Different source distributions of the wave equation correspond to different free-field radiation patterns. To resolve these sources adequately, there is an upper limit in coarsening the mesh. Therefore, the advantage of a hybrid aeroacoustic workflow is that the source field on the CFD mesh can be kept the same while investigating different resolutions of the CAA grid in the source region. Quantitative plots of the analytic source fields on the CFD mesh are shown in Fig. 5.

Table 2. Two-dimensional analytic source fields.

Monopole source	Dipole source	Quadrupole source
$f_{\mathrm{MP}} = \frac{1}{\sqrt{2\pi\sigma^2}}\, e^{-\frac{x^2+y^2}{2\sigma^2}}$	$f_{\mathrm{DP}} = \frac{-x}{\sqrt{2\pi\sigma^2}\cdot\sigma^2}\, e^{-\frac{x^2+y^2}{2\sigma^2}}$	$f_{\mathrm{QP}} = \frac{xy}{\sqrt{2\pi\sigma^2}\cdot\sigma^4}\, e^{-\frac{x^2+y^2}{2\sigma^2}}$



(a) Monopole source. (b) Dipole source. (c) Quadrupole source.

Fig. 5. Depiction of the source regions from Fig. 6, with different generic sources used for the validation of the interpolation algorithm. Absolute value of the source strength is displayed.

The propagation region does not differ, except in the surrounding area of the source, for a smooth transition between the grid of the source region and the propagation region. The complete computation domain consists of a circular source region with a radius of 4.95 m, a propagation region of 240 m × 240 m, and a perfectly matched layer (PML) region with a width of 10 m to account for the free radiation condition (see Fig. 6). We performed a time-harmonic simulation at a frequency of $f_{\text{excite}} = 30\,\text{Hz}$ and ambient conditions ($T = 293.15\,\text{K}$), resulting in a wavelength of $\lambda = 11.4\,\text{m}$. The element size inside the propagation region resolves the acoustic wave by 20 linear elements per wavelength. All used meshes are unstructured meshes. The propagation domain discretizes the acoustic equations to resolve wave propagation correctly. The cell size of the source mesh is chosen such that it is half of the cell sizes of the finest coarser target mesh and is also used for the reference mesh. The acoustic source fields are prescribed on the source region of the CFD mesh (see Fig. 7(a)). Now, the acoustic source field is interpolated from the source mesh inside the source region on the acoustic target mesh. Then, the interpolated acoustic source field on the acoustic

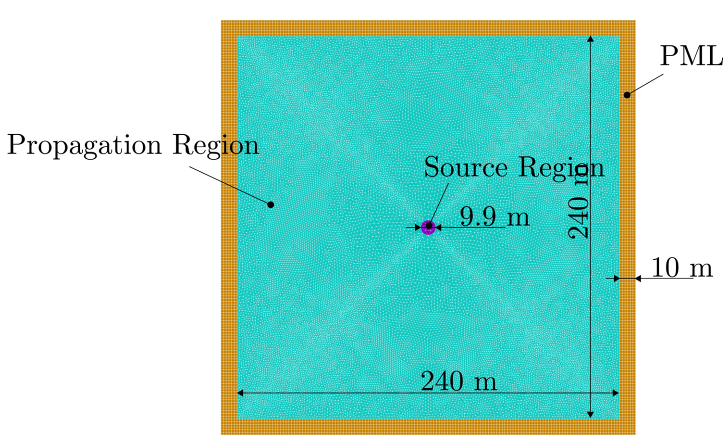
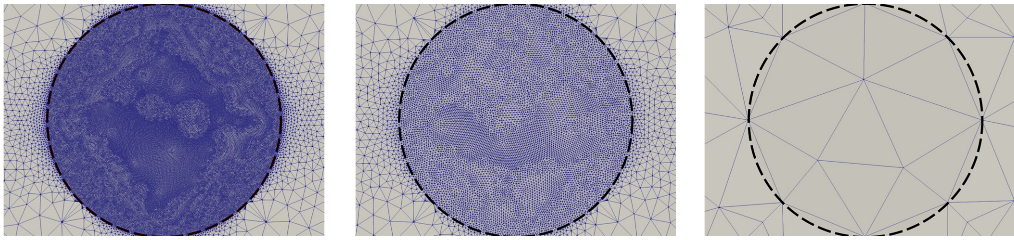


Fig. 6. Depiction of the regions and the mesh of the CAA simulation.



(a) Reference mesh. (b) Mesh 1. (c) Mesh 7.

Fig. 7. Different meshes for the interpolations, with a very fine mesh of the analytic source, and the finest and coarsest target mesh.

target mesh is used as a right-hand side for an acoustic propagation simulation. The acoustic computation is solved time-harmonically.

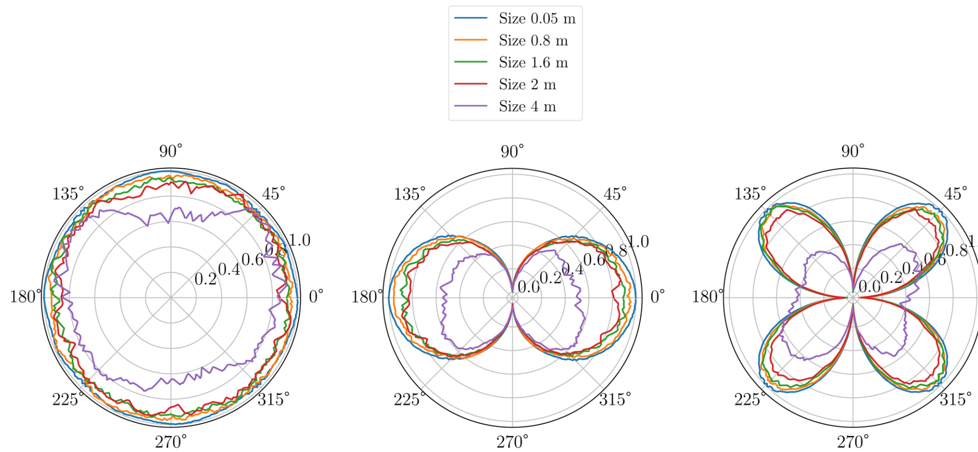
For the numerical investigation, different mesh resolutions are used as acoustic target meshes (see Fig. 7). The black dashed circle marks the source region from Fig. 6. The prescribed right-hand side values of this region are plotted in Fig. 5. The different mesh sizes of the source domains used for this verification are listed in Table 3. The effects of the radiation are discussed below. The last column of this table describes the relation of the element size to the spatial parameter σ of the analytic sources. As the spatial parameter σ represents the size of the source, this column describes the relation between the element and the source size.

The evaluation of the acoustic radiation for the three different source fields is shown in Fig. 8. For each source-setup, for a circular microphone array with a radius of 20 m, the resulting acoustic pressure relative to the maximum evaluated pressure is plotted.

The acoustic result is almost identical for mesh sizes up to 0.8 m, or to a ratio of the element size to spatial parameter σ of about 1, especially for monopole sources. For larger ratios of mesh size to σ, the radiation patterns are represented correctly; however, the amplitude clearly shows deviations from the real value. The interpolation only fails for the

Table 3. Different meshes used in the source region to investigate the cut-volume cell interpolation.

	Number of cells	Number of nodes	Element size in m	Element size/σ
Mesh -2	1 267 140	634 815	0.0125	$0.01\bar{6}$
Mesh -1	305 494	153 370	0.025	$0.0\bar{3}$
Reference mesh	77 416	39 021	0.05	$0.0\bar{6}$
Mesh 1	18 378	9346	0.1	$0.1\bar{3}$
Mesh 2	4626	2392	0.2	$0.2\bar{6}$
Mesh 3	1160	621	0.4	$0.5\bar{3}$
Mesh 4	292	167	0.8	$1.0\bar{6}$
Mesh 5	82	52	1.6	$2.1\bar{3}$
Mesh 6	56	37	2	$2.\bar{6}$
Mesh 7	12	11	4	$5.\bar{3}$



(a) Monopole radiation. (b) Dipole radiation. (c) Quadrupole radiation.

Fig. 8. Results of the coarsest discretizations for the generic sources (cut-volume cell). We omitted the results of the finer discretizations since the lines only improve slightly compared to size 0.05 m.

coarsest mesh with a cell size of 4 m. For the setups where the CAA meshes are finer than the CFD mesh, the deviations are very similar to the finest coarser mesh, which can be verified in Table 4. From these results, we conclude that the cut-volume cell approach can be applied to a large range of acoustic mesh sizes. The upper limit of the acoustic mesh size is given by the source size, which is the spatial parameter σ in this verification example.

The results of the cell-centroid approach are almost identical to the cut-volume cell, for all considered element size to σ ratios, as shown in Table 4.

Hypothesis 4. To conclude for compact aeroacoustic sources, it is essential that the source is at least discretized by one element regarding the decay parameter σ. Overall, this discretization criterion slightly depends on the polar source characteristic (see Table 4).

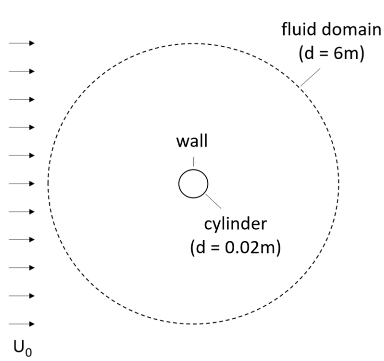
Table 4. Averaged deviations in the acoustic results between reference mesh and finer or coarser meshes (concerning the reference mesh in percent), using cell-centroid and cut-volume cell interpolation.

	Cell-centroid			Cut-volume cell		
	Monopole	Dipole	Quadrupole	Monopole	Dipole	Quadrupole
Mesh-2	−0.45	1.74	0.12	−0.45	1.75	0.13
Mesh-1	−0.36	1.84	0.30	−0.36	1.84	0.29
Mesh 1	−0.23	1.27	0.46	−0.23	1.27	0.46
Mesh 2	−0.21	1.55	0.73	−0.20	1.56	0.73
Mesh 3	0.56	2.53	2.25	0.56	2.53	2.25
Mesh 4	2.58	5.32	4.69	2.58	5.33	4.69
Mesh 5	6.03	12.58	8.61	6.04	12.60	8.62
Mesh 6	8.04	12.98	14.39	8.05	12.98	14.39
Mesh 7	27.21	47.63	70.58	27.25	47.67	70.67

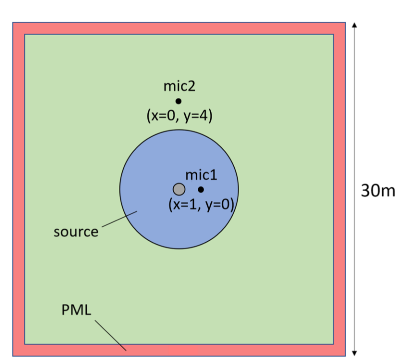
However, the interpolation error of the dipole and quadrupole source shapes are higher, using the same discretization. Following these rules, acoustic meshes can be designed inside the aeroacoustic source region. This estimation can be incorporated into a post-processing routine of the CFD simulation that calculates the values for the source distribution σ.

4. Application — Cylinder in a Crossflow

The first validation example demonstrates the impact of the different interpolation strategies on the near-field and far-field of acoustic sound propagation. We choose the 2D, laminar flow over a cylinder (Re $= 200$, Ma $= U_0/c = 0.03$, see Fig. 9) to report the impact. An unsteady flow field is observed, with a periodic vortex shedding at a Strouhal number of 0.205 and 0.41.[24] This periodic fluid instability inside the wake of this cylinder generates tonal components in the sound field. The CFD was computed employing *Ansys Fluent* (ANSYS, Southpointe, PA, USA). The incompressible pressure was used to compute the aeroacoustic sources based on (2). Afterward, this source term was conservatively interpolated to the acoustic mesh by the two interpolation schemes (see Secs. 2.1 and 2.2). Having obtained the sources, the acoustic propagation was computed and the results were compared systematically.[4,25] We compare four acoustic source domain meshes. Table 5 sums up the investigated meshes ranging from refined acoustics meshes to coarse acoustic meshes. The reference mesh is the mesh used during the CFD simulation. Time-harmonic acoustic simulations are carried out, using *openCFS*.[26] Regarding the interpolation method, the results are compared in terms of amplitude spectra at microphone positions 1 and 2, the directivity plot in the near-field and far-field, the source directivity in the near-field and the phase of the acoustic field in the near-field. *A priori*, we expect good agreement of both interpolations approaches in the far-field whereas the results in the source region are expected to differ slightly. The different acoustic meshes used for the mesh dependency study of the interpolation in 2D are summarized in Table 5. As the simulation reference, we use the CFD grid and compute the aeroacoustic sources there using numerical 2D Lagrange integration on the grid; using the same mesh for CAA and CFD, the cell-centroid and the cut-volume cell



(a) CFD setup.



(b) CAA domain.

Fig. 9. Schematic of the CFD setup and the CAA domain of the cylinder in a crossflow.

Table 5. CAA mesh specifications and computational time of the source domain mesh study.

Mesh	Description	Number of elements	Interpolation effect obtained	Setup time	Execution per interpolation
Reference	CFD mesh	531 864	—	—	0.51 s
Mesh 1	CFD mesh (refined)	320 800	Chequerboard near cylinder	530 s	0.48 s
Mesh 2	CAA mesh	20 200	Good resolution of acoustics	217 s	0.20 s
Mesh 3	Coarse CAA mesh	420	Poor acoustics	71 s	0.18 s

algorithm (as explained in Secs. 2.1 and 2.2) are identical. Considering Mesh 1, we obtain a chequerboard source pattern when using the cell-centroid interpolation. Mesh 1 is locally refined around the cylinder to illustrate the theoretically investigated effects for $\Gamma > 1$ (see Fig. 4). Mesh 2 adequately resolves the aeroacoustic source domain (20 elements per σ) and the acoustic propagation (20 elements per λ). Mesh 3 under-resolves the aeroacoustic source domain (about one element per 2σ) and the acoustic propagation (20 elements per λ). We expect deviations from the results for Mesh 3. Due to the difference of these two interpolations, we distinguish between the setup time and the execution time of the interpolation, considering a single field interpolation (one frequency step for time-harmonic simulations). Table 5 reports the setup and execution times, respectively. It is important to note that the setup time is a function of the possible intersections, which increases with the number of target elements. However, this intersection algorithm is specific for the dimensionality of the domain. We found that the intersection algorithm in 2D depends on the possible intersection candidates strongly. However, the execution time per interpolation is primarily dominated by the file output performance of the file system, leading to negligible performance increase for Mesh 3.

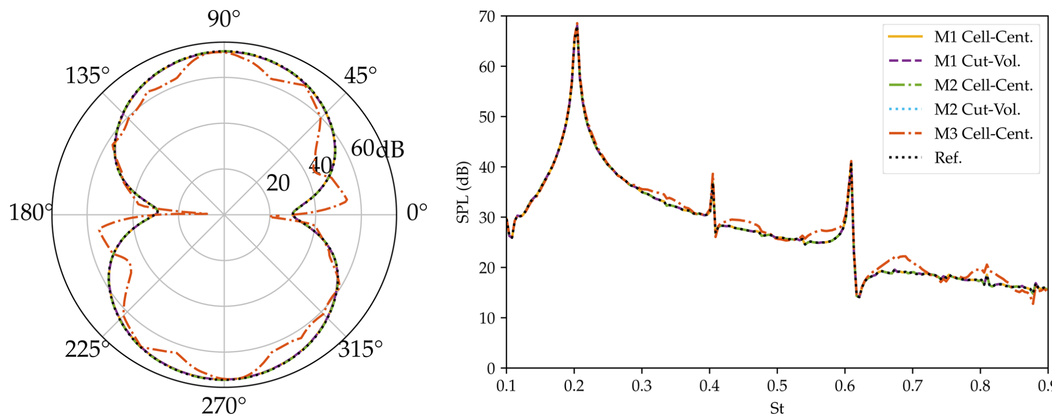
4.1. *Comparison of methods*

First, we compare the spectra of the sound pressure levels to investigate the impact of different interpolation schemes in combination with different meshes. Figure 10 shows the spectra and the directivity for all combinations, respectively. The spectra of the far-field microphone and the directivity reveal that all interpolations and mesh combinations, except Mesh 3 perform well. However, considering the relative number of elements inside the source region of Mesh 3 to the CFD mesh (see Table 5), the number of elements is reduced by a factor of 1000. Despite the strong reduction of the mesh size, the result of Mesh 3 is remarkable when considering the sound pressure level at the microphone positions only. Both interpolation strategies perform well for far-field characteristics.

To quantify convergence, we use the frequency averaged sound pressure level

$$\bar{L}_{\mathrm{pf}} = 10 \log \left(\frac{1}{N} \sum (p^{\mathrm{a}})^2 / p_0^2 \right) \mathrm{dB}, \tag{16}$$

with the acoustic reference pressure $p_0 = 20\,\mu\mathrm{Pa}$. Table 6 shows the quantitative evaluation of the sound pressure spectra for two microphone positions. In 2D and for the acoustic



(a) Array position $r = 2\,\mathrm{m}$.

(b) Far-field microphone at (0.0 m and 4.0 m).

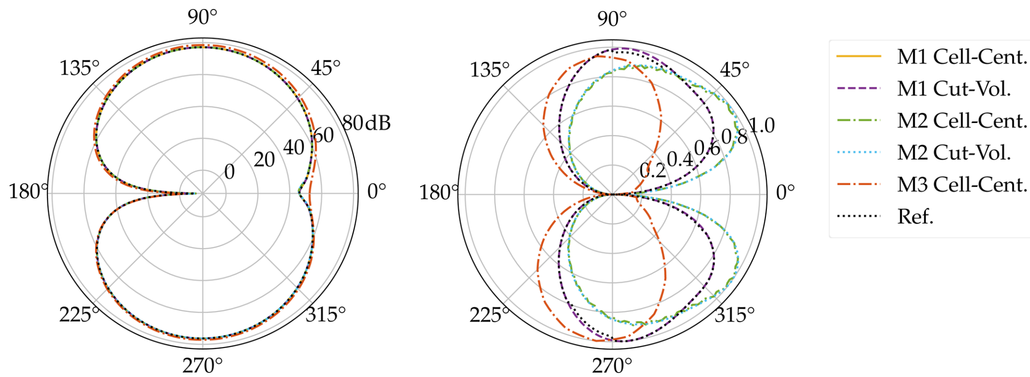
Fig. 10. Far-field directivity pattern of radiated sound and the spectra of the sound pressure level at microphone location in the far-field. Both methods yield comparable results for the different meshes used.

far-field (Mic2), monotonic convergence is obtained for both interpolation algorithms. However, the microphone position inside the wake of the cylinder shows that Mesh 1 results in lower average amplitudes, which can be attributed to the higher resolution inside the source domain leading to errors.

Next, we will investigate the near-field properties of the source terms in the frequency domain and the near-field properties of the pressure. Figure 11 shows the directivity plot of the sound pressure spectra and the FE sources in the acoustic near-field of the cylinder. Again, the sound radiation yields similar results for the meshes except for Mesh 3 (see Fig. 11(a)). However, the nodal FE loads are differing significantly due to the different areas of influence of each element. Since most sources occur close to the cylinder, Mesh 2 features a higher amplitude in the downstream direction. The very coarse Mesh 3 averages over the vortex street, resulting in a lower downstream amplitude. Additionally, we notice that Mesh 1 in combination with the cell-centroid interpolation has no contributions at $r = 0.0101\,\mathrm{m}$. Overall, we see large deviations of the sources in the near-field but with minor influence on the sound radiation.

Table 6. Frequency averaged sound pressure level ($\bar{L}_{\mathrm{pf}}$ in dB) for the different evaluated combinations.

	Cell-centroid Mic1	Cut-volume cell Mic1	Cell-centroid Mic2	Cut-volume cell Mic2
Reference	45.8200	45.8200	47.6539	47.6539
Mesh 1	45.7625	45.7625	47.6542	47.6542
Mesh 2	45.8208	45.8207	47.6555	47.6554
Mesh 3	46.0248	46.0246	48.1726	48.1724



(a) Array position $r = 0.1\,\mathrm{m}$. (b) Array position $r = 0.0101\,\mathrm{m}$.

Fig. 11. Directivity plot of the sound pressure spectra and the FE sources in the acoustic near-field of the cylinder.

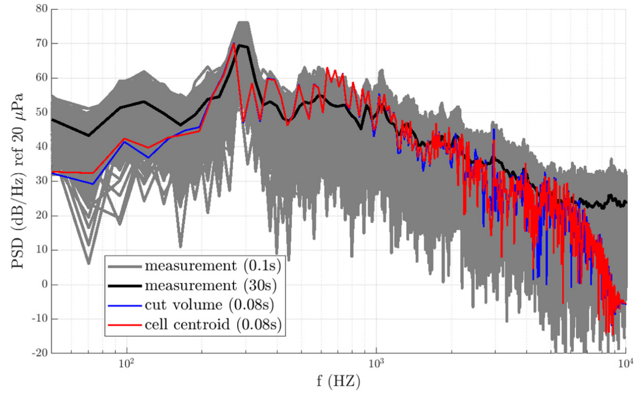
4.2. *Summary — cylinder in a crossflow*

Both methods yield comparable results in the acoustic far-field, although the FE sources look odd for the cell-centroid interpolation and the refined source mesh. This fact has been explained theoretically in the previous section and heuristically proven for 2D in this section. Both the computational time of the CAA simulation and the stored data decreased after the reduction of the mesh size inside the source region. The stored CFD data in terms of source terms can be decreased by 96.6%, comparing the reference mesh storage consumption to Mesh 2. Regarding Hypothesis 3, we have compact acoustic sources with a Helmholtz number of 0.01 and a compact source re-distribution. Thus, the chequerboard-like source distribution of Mesh 1 does not influence the sound propagation.

5. Application — Axial Fan

The mentioned methods are applied to a validated fan simulation[17] (see Fig. 12(a)). It has a diameter of $0.5\,\mathrm{m}$, consists of nine flat plates, is operated at $1500\,\mathrm{rpm}$ with a volume flow of $1.3\,\mathrm{m}^3/\mathrm{s}$, and is mounted in a duct together with the motor. A detached eddy CFD simulation provided the incompressible pressure to compute the source terms according to Eq. (2). The source terms are computed on the CFD mesh ($33\,\mathrm{M}$ cells, with $22\,\mathrm{M}$ cells in the rotating region and a maximum cell size of $1\,\mathrm{mm}$) and then interpolated to the acoustic domain. The details of the setup are given a previous study.[17] For the two different interpolation strategies, we performed a grid convergence study within the rotating source region, including five meshes. Inside the large stationary propagation region, we have a large mesh ratio Γ resulting in source artifacts (see Fig. 1). But as explained, we expect nearly no effect on the resulting sound pressure level at the microphone positions. The acoustic free field is modeled by a PML and the nozzle, duct, strut, driving shaft and motor are modeled as sound-hard walls. After the interpolation, the acoustic propagation computation was performed with openCFS.[26]

(a) Geometry of the investigated fan.



(b) Comparison of the PSD using the cell-centroid-based and the cut-volume cell interpolation algorithm with measurement results.

Fig. 12. Investigated fan and the comparison of the interpolation method.

5.1. *Comparison of the interpolation algorithm*

The source term was interpolated to the acoustic mesh using the two different interpolation approaches. The power spectral density (PSD) results of the two simulations are compared with measurement results in Fig. 12(b) at a position of 1 m in front of the inlet nozzle. The black line represents a measurement with a measurement time of 30 s. The gray lines are measurement results with a measurement time of 0.1 s, which is close to the simulation time of 0.08 s. In the low-frequency range, the simulations underestimate the measured PSD, which might be due to the short numerical simulation time, the aeroacoustic formulation, and the cut-off frequency of the anechoic room. Both simulation results have a steep drop above 6 kHz where the mesh resolution in the propagation domain becomes coarse compared to the acoustic wavelength. Therefore, the respective wavelengths cannot be resolved anymore, leading to numerical damping of the acoustic waves by the time-stepping scheme with controlled dispersion (Hilber–Hughes–Taylor). The dominant peak at a frequency of 300 Hz is a result of both the interaction of the tip gap flow and the blade passing frequency.[27] It should be emphasized that these frequencies are not the same; however, due to the frequency resolution used in this case, they merge to one peak. This peak is represented in both frequency and amplitude. The red curve shows the result of the cell-centroid-based approach, while the blue curve represents the results of the cut-volume cell approach. Both interpolation methods result in a good prediction of the PSD in a wide frequency range and therefore are usable in cases with complex geometry and distorted grid cells of varying cell sizes in real applications. This is in agreement with the previous findings considering Hypotheses 1 and 3. The intersection algorithm has an additional computation time of about 45 min and heavily depends on the number of CFD cells. Table 7 shows a small increase in the intersection time with increasing CAA elements.

Table 7. Different meshes used in the source region to investigate the cut-volume cell interpolation and the cell-centroid-based interpolation. The coverage ratio measures the relative coverage of CAA cells by CFD cells for the cell-centroid-based interpolation. The intersection time describes the excess CPU time of the cut-volume cell interpolation, due to the intersection algorithm. The execution time describes the averaged CPU time (using four threads), that was used to calculate the acoustic sources with the cut-volume interpolation for one time step.

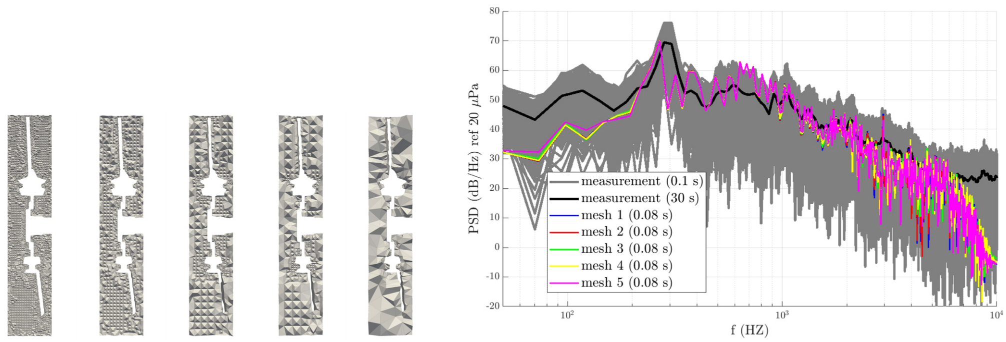
	Source region elements	Source region nodes	Max. element size	Coverage ratio ε	Intersection time	Execution time per step
Mesh 1	7 690 908	1 322 937	7 mm	85.24%	49 min	33.3 s
Mesh 2	1 021 697	181 186	12 mm	99.78%	45 min	29.0 s
Mesh 3	555 562	96 535	18 mm	92.87%	42 min	24.7 s
Mesh 4	337 013	58 708	24 mm	95.14%	41 min	23.9 s
Mesh 5	153 256	27 765	48 mm	98.36 %	42 min	25.3 s

5.2. *Artifacts inside the propagation domain*

As stated previously, in regions with a strongly refined CAA mesh (large mesh ratio Γ), artifacts occur inside the FE source distribution for the cell-centroid interpolation. Figure 1 shows these artifacts and compares them to the cut-volume cell interpolation, having no artifacts in the acoustic sources. The results of Fig. 12(b), the theoretical considerations, and the 2D simulations demonstrated that these artifacts introduce a small phase error $\Delta\varphi_{\mathrm{error}}$ with minor impact on the acoustic results of this application. However, if these artifacts are in regions with large source terms they might have an impact. Alternatively, source blending can be utilized to mitigate spurious acoustic modes.[28,29]

5.3. *Mesh dependency*

The same setup as before was used to investigate the influence of the CAA grid resolution on the cut-volume cell algorithm. This time, the cut-volume cell algorithm was used to interpolate the acoustic source on five different grids in the rotating region. All grids are tetrahedral and have different grid sizes. The numbers of elements in the source region are shown in Table 7, where Mesh 1 is the finest and Mesh 5 is the coarsest mesh with comparably bad mesh quality. The mesh of the propagation domain was not changed for the different simulations and counted 1 804 377 nodes, since altering this mesh would add additional deviation of the results that are not subject to this mesh study. The meshes of the rotating region are displayed in Fig. 13(a), where the meshes become coarser from left to right. The minimum cell size of the CAA mesh occurs in the tip gap, where the size is 1 mm for all CAA meshes. This is necessary to resolve the gap with at least two elements, independently of the discretization of the rest of the mesh. Therefore, the elements in the tip gap dominate the total number of elements for coarse meshes and therefore limit the



(a) Different meshes used for the computation

(b) Acoustic PSD for different discretizations

Fig. 13. Comparison of different discretizations (from the finest Mesh 1 to the coarsest Mesh 5) using the cut-volume cell-based interpolation.

minimal number of elements. The maximum CFD cell size in this location is 0.2 mm, which yields a ratio of $\Gamma = 0.2$.

The PSD that resulted from the CAA computation using the interpolated source term and the PSD of the measurements are displayed in Fig. 13(b). Mesh 1 to 4 deliver almost identical results up to a frequency of 1 kHz. Above this frequency, it can be seen that the finer meshes lead to more distinctive peaks, but the deviation is rather small. Mesh 5 yields small deviations also for low frequencies. This could be caused by the rather poor mesh quality due to the limitation in maximum mesh size in the tip gap mentioned above. The deviations in the high-frequency range are assumed to have its origin in the mesh density of the source region, where the coarsest mesh meets its limitations to resolve the geometric shape of the source terms. These results verify the robustness of the two conservative interpolation algorithms in 3D. Furthermore, the reduction of the nodes in the rotation region decreases the overall number of nodes. This reduces the size of the FE system and, therefore, the computational time. For the finest mesh, the total CPU time was 930 h and for the coarsest, the total CPU time was reduced to 229 h. The strong reduction in CPU time is possible since the FE simulation time is proportional to the square of the number of nodes. Overall, in this axial fan case due to geometrical properties and the properties of the aeroacoustic source, being a result of the tip gap flow, limits the breakdown of the cut-volume cell approach, since the tip gap must be resolved at least by one element.

For further comparisons, the averaged sound pressure level is introduced to quantify the accuracy of each simulation. The averaged sound pressure level is defined by

$$\bar{L}_{\mathrm{P}} = 10 \log \left(\frac{1}{T} \int p_n^2 \mathrm{d}t / p_0^2 \right) \mathrm{dB}, \tag{17}$$

with a reference pressure $p_0 = 20\,\mu\mathrm{Pa}$. Table 8 shows the averaged sound pressure level at the microphone position for all investigated simulations. The sound pressure level only varies slightly for the different meshes and both simulation setups. Furthermore, the relative

Table 8. Comparison of the sound pressure level and the deviation of the PSD for the different meshes using cut-volume and cell-centroid interpolation.

	Cut-volume Cell interpolation		Cell-centroid-based interpolation		
	$e_{\mathrm{RMS}}{}^{*}$	$\bar{L}_{\mathrm{P}}$	$e_{\mathrm{RMS}}{}^{*}$	$e_{\mathrm{RMS}}{}^{**}$	$\bar{L}_{\mathrm{P}}$
Mesh 1	0%	88.2972 dB	5.45%	0%	88.2835 dB
Mesh 2	3.37%	88.2550 dB	6.52%	5.45%	88.2454 dB
Mesh 3	7.82%	88.1963 dB	8.85%	6.38%	88.1748 dB
Mesh 4	10.83%	88.1617 dB	10.81%	8.70%	88.1490 dB
Mesh 5	13.85%	88.0037 dB	13.85%	12.66%	88.0038 dB

Note: *In reference to simulation of Mesh 1 and cut-volume cell interpolation.
**In reference to simulation of Mesh 1 and cell-centroid-based interpolation.

root mean squared error of the PSD is calculated by

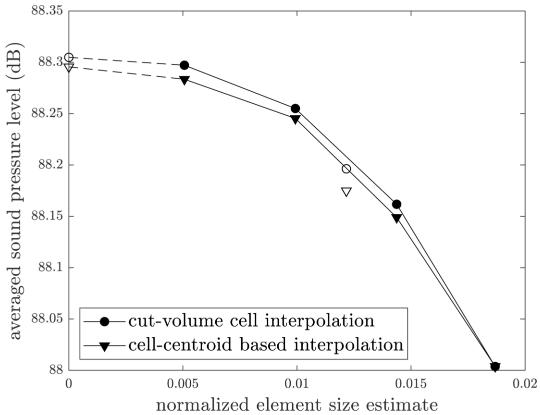
$$e_{\mathrm{RMS}} = \sqrt{\frac{\sum_{f_{\min}}^{f_{\max}} (\mathrm{PSD}_{\mathrm{ref}} - \mathrm{PSD})^2}{\sum_{f_{\min}}^{f_{\max}} \mathrm{PSD}_{\mathrm{ref}}^2}} \tag{18}$$

and depicted in Table 8 with reference to the finest mesh (Mesh 1). $f_{\min}$ and $f_{\max}$ are chosen to be 100 Hz and 5 kHz, respectively. Both interpolation methods yield small deviations for the finest mesh. With increasing mesh size the error e_{RMS} increases, consistently. However, both conservative interpolation methods yield similar and acceptable accuracy.

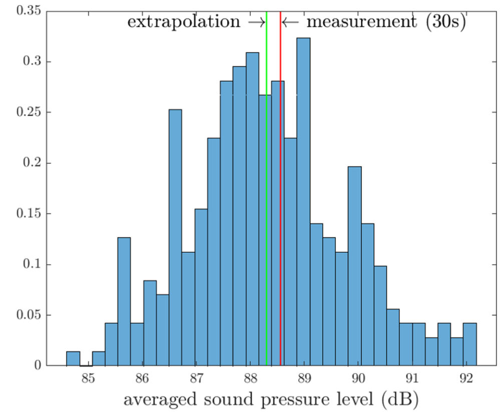
Another aspect is the monotonically converging averaged sound pressure level for finer meshes. To further analyze this convergence behavior, the grid convergence index (GCI)[30] is introduced. It gives an estimate for the uncertainty of a result variable concerning the mesh resolution. Therefore, a Richardson extrapolation is used for the averaged SPL for Meshes 1, 2, 4 and 5 in order to estimate an approximate result for infinite mesh resolution. Mesh 3 is not included in this analysis as it is seen to be an outlier as well as to guarantee a refinement ratio $r_{i,j} = \frac{h_j}{h_i} > 1.3$, as proposed by Ref. 31. The relative refinement is estimated by the ratio of the averaged element sizes $r_{i,j} \approx \sqrt[3]{\frac{n_{\mathrm{el},i}}{n_{\mathrm{el},j}}}$. As a result of this analysis, estimated averaged SPL $\bar{L}_{\mathrm{p}}^{\mathrm{ext}} = 88.2972\,\mathrm{dB}$ and $\bar{L}_{\mathrm{p}}^{\mathrm{ext}} = 88.2835\,\mathrm{dB}$ and orders of convergence $p = 2.835$ and $p = 2.659$ are extrapolated for the cut volume and cell centroid interpolated simulations, respectively. The GCI is defined by

$$\begin{aligned} \mathrm{GCI}_i^{\mathrm{fine}} &= F_{\mathrm{s}} \left| \frac{\varepsilon_L}{1 - r_i^p} \right|, \\ \mathrm{GCI}_i^{\mathrm{coarse}} &= F_{\mathrm{s}} \left| \frac{r_i^p \varepsilon_L}{1 - r_i^p} \right| \end{aligned} \tag{19}$$

with the relative deviation $\varepsilon_L = \frac{\bar{L}_{\mathrm{P}}^{\mathrm{coarse}} - \bar{L}_{\mathrm{P}}^{\mathrm{fine}}}{\bar{L}_{\mathrm{P}}^{\mathrm{fine}}}$ and a recommended safety factor $F_{\mathrm{s}} = 1.25$ (according to Ref. 30). Figure 14(a) shows the averaged sound pressure level in respect to the normalized element size estimate and the extrapolated value. Table 9 summarizes the results



(a) Grid convergence regarding mesh refinement using Richardson extrapolation.



(b) Averaged sound pressure level from the measurement.

Fig. 14. Convergence of averaged sound pressure level and measurement results.

Table 9. GCI based on Richardson extrapolation and measurement results.

	Cut-volume Cell interpolation		Cell-centroid-based interpolation		Refinement ratio
	$\bar{L}_{\mathrm{P}}$	GCI (10^{-3})	$\bar{L}_{\mathrm{P}}$	GCI (10^{-3})	$r_{i,i+1}$
Measurement (30 s)	88.556* dB	—	88.556* dB	—	—
Extrapolation	88.3049 dB	—	88.2956 dB	—	—
Mesh 1	88.2972 dB	0.1041	88.2835 dB	0.1084	1.960
Mesh 2	88.2550 dB	0.7142	88.2454 dB	0.8172	1.447
Mesh 4	88.1617 dB	2.0301	88.1490 dB	2.0398	1.3004
Mesh 5	88.0037 dB	4.2745	88.0038 dB	4.1013	—

Note: *For 30 s measurement, [84.746, 92.176] dB bandwidth for measurements with 0.1s measurement time.

for the cut-volume interpolation and includes the averaged SPL for the 30 s measurement as well as the spread for 0.1 s measurements in the frequency range of 100 Hz to 5 kHz. The measurements yield an averaged SPL of $\bar{L}_{\mathrm{P}} = 88.556\,\mathrm{dB}$ for the 30 s measurement. Figure 14(b) also shows the SPL spread for 0.1 s measurement data. The measured signal generally give a higher averaged sound pressure level because low and high frequencies are underestimated by the simulation. However, all analyzed meshes yield an approximation for the averaged SPL that lies well inside the 0.1 s measurement bandwidth.

To conclude, both methods yield comparable results with beneficial convergence properties.

6. Conclusions

Both interpolation methods, the cell-centroid-based interpolation method and a cut-volume cell approach, are explained, discussed in detail, and analyzed toward its applicability regarding four hypotheses. Thereafter, both methods are applied to the simulations of an axial fan in 3D and a cylinder in a crossflow in 2D to further proof the hypotheses. The cut-volume cell approach takes the geometry of the interpolation cell into account and therefore improves robustness for meshes with skewed cells and different cell sizes, especially when the CAA grid becomes smaller than the CFD grid. However, the smoother FE source field of the cut-volume cell approach for large mesh ratios results only in partial benefits for the acoustic propagation accuracy. This fact was derived, systematically. An upper bound for the CAA cell size without changing the radiation characteristics of noise sources was presented. Comparing the two interpolation methods, we show how energy is conserved or transferred from the interpolated source mode to artificial modes. Therefore, a measure of energy conservation is developed by the coverage ratio. This coverage ratio computes the ratio of CAA cells that are covered by a CFD cell and the total number of CAA cells for the cell-centroid interpolation. Based on this simple mesh metric, we can quantify the quality of a simple source term calculation procedure and switch to a more sophisticated algorithm if necessary. Table 10 summarizes the application range for different mesh ratios.

The application of both methods to the cylinder example in 2D yields comparable results in the acoustic far-field, despite that the FE sources look odd for the cell-centroid interpolation and the refined mesh. The mesh convergence study showed that the computational time of the CAA simulation and the stored data (96.6%) decreased significantly.

Both methods predict the PSD results, for the application to an axial fan configuration in 3D, well. But for the finest mesh, the coverage ratio decreases which indicates a possible error source for the cell-centroid-based interpolation. The robustness of the interpolation methods was shown for different mesh ratios and assessed by a grid convergence study. Based on the averaged sound pressure level, both interpolation methods show a monotonic convergence. Additionally, the acoustic result changed very slightly even for a strong reduction of elements in the source region (factor of 50). Overall, the number of unknowns dropped by a factor of 1.71. The CAA simulation's computational time was reduced by a factor of 4.06 for the

Table 10. Application summary of cell-centroid-based procedure and the cut-volume cell procedure.

	Cell-centroid	Cut-volume cell
CAA elements *larger* than CFD cells **or** source shift$\ll \lambda$	*Recommended* (Typical Hybrid CAA)	Computational too expensive
CAA elements *smaller* than CFD cells **and** source shift approaching λ	Inaccurate	*Recommended* (Special cases CAA)

coarsest mesh. Finally, the validated simulation results showed good accordance in both PSD and averaged sound pressure level. Leading to the conclusion, the presented interpolation algorithms are robust and accurate for typical low Mach number aeroacoustic applications.

Acknowledgments

The authors acknowledge support from the German Research Foundation (DFG) under DO 1247/10-1 (no. 391215328), the Austrian Research Council (FWF) under no. I 3702 and the Austria Research Promotion Agency (FFG) under no. 867971.

References

1. S. Schoder and M. Kaltenbacher, Hybrid aeroacoustic computations: State of art and new achievements, *J. Theor. Comput. Acoust.* **27**(4) (2019) 1950020.
2. D. Crighton, Computational aeroacoustics for low Mach number flows, in *Computational Aeroacoustics* (Springer, 1993), pp. 50–68.
3. C. Wagner, T. Hüttl and P. Sagaut (eds.), *Large-Eddy Simulation for Acoustics* (Cambridge University Press, 2007).
4. S. Caro, Y. Detandt, J. Manera, F. Mendonca and R. Toppinga, Validation of a new hybrid CAA strategy and application to the noise generated by a flap in a simplified HVAC duct, *15th AIAA/CEAS Aeroacoustics Conf.*, 2009.
5. T. Schröder, P. Silkeit and O. Estorff, Influence of source term interpolation on hybrid computational aeroacoustics in finite volumes, *InterNoise 2016*, 2016, pp. 1598–1608.
6. P. E. Farrell and J. R. Maddison, Conservative interpolation between volume meshes by local Galerkin projection, *Comput. Methods Appl. Mech. Eng.* **200** (2011) 89–100, doi:10.1016/j.cma.2010.07.015.
7. I. Steinbrecher, M. Mayr, M. J. Grill, J. Kremheller, C. Meier and A. Popp, A mortar-type finite element approach for embedding 1D beams into 3D solid volumes, in *Computational Mechanics* (Springer, 2020), pp. 1–22.
8. C. Carton, L. Georges, P. Geuzaine, Y. Detandt and S. Caro, Further analysis of a hybrid CAA method based on Argo and Actran/LA: Case of the Helmholtz resonator, *15th AIAA/CEAS Aeroacoustics Conf.*, No. 3115, 2009.
9. J. Manera, Y. Detandt, D. D'Udekem and S. Detry, Aero-acoustic predictions of automotive instrument panel ducts, *SAE NVH Conf.*, SAE 09NVC-0131, St. Charles, IL, 2009.
10. M. Piellard and C. Bailly, A hybrid method for computational aeroacoustic applied to internal flows, *NAG-DAGA Int. Conf. Acoustics*, 2009.
11. R. L. Hardy, Multiquadric equations of topography and other irregular surfaces, *J. Geophys. Res.* **76**(8) (1971) 1905–1915.
12. R. L. Hardy, Theory and applications of the multiquadric-biharmonic method 20 years of discovery 1968–1988, *Comput. Math. Appl.* **19**(8–9) (1990) 163–208.
13. T. Rendall and C. Allen, Unified fluid–structure interpolation and mesh motion using radial basis functions, *Int. J. Numer. Methods Eng.* **74**(10) (2008) 1519–1559.
14. M. Cordero-Gracia, M. Gómez and E. Valero, A radial basis function algorithm for simplified fluid-structure data transfer, *Int. J. Numer. Methods Eng.* **99**(12) (2014) 888–905.
15. S. Schoder, K. Roppert, M. Weitz, C. Junger and M. Kaltenbacher, Aeroacoustic source term computation based on radial basis function, *Int. J. Numer. Methods Eng.* **121** (2020) 2051–2067, doi:10.1002/nme.6298.

16. M. Kaltenbacher, M. Escobar, I. Ali and S. Becker, Numerical simulation of flow-induced noise using LES/SAS and Lighthill's acoustics analogy, *Int. J. Numer. Methods Fluids* **63**(9) (2010) 1103–1122.
17. M. Kaltenbacher, A. Hüppe, A. Reppenhagen, F. Zenger and S. Becker, Computational aeroacoustics for rotating systems with application to an axial fan, *AIAA J.* **55**(11) (2017) 3831–3838.
18. P. Croaker, N. Kessissoglou, R. Kinns and S. Marburg, Fast low-storage method for evaluating Lighthills volume quadrupoles, *AIAA J.* **51**(4) (2013) 867–884.
19. J. Hardin and D. Pope, An acoustic/viscous splitting technique for computational aeroacoustics, *Theor. Comput. Fluid Dyn.* **6**(5–6) (1994) 323–340.
20. W. Z. Shen and J. N. Sorensen, Aeroacoustic modelling of low-speed flows, *Theor. Comput. Fluid Dyn.* **13**(4) (1999) 271–289.
21. R. Ewert and W. Schröder, Acoustic perturbation equations based on flow decomposition via source filtering, *J. Comput. Phys.* **188**(2) (2003) 365–398.
22. J. Seo and Y. J. Moon, Perturbed compressible equations for aeroacoustic noise prediction at low mach numbers, *AIAA J.* **43**(8) (2005) 1716–1724.
23. C.-D. Munz, M. Dumbser and S. Roller, Linearized acoustic perturbation equations for low Mach number flow with variable density and temperature, *J. Comput. Phys.* **224**(1) (2007) 352–364.
24. M. M. Zdravkovich, *Flow Around Circular Cylinders: Volume 2: Applications* (Oxford University Press, 1997).
25. A. Hüppe and M. Kaltenbacher, Comparison of source term formulations for computational aeroacoustics, *19th AIAA/CEAS Aeroacoustics Conf.*, No. 2170, 2013.
26. M. Kaltenbacher, *Numerical Simulation of Mechatronic Sensors and Actuators: Finite Elements for Computational Multiphysics* (Springer, Berlin, Heidelberg, 2015).
27. F. Krömer, J. Müller and S. Becker, Investigation of aeroacoustic properties of low pressure axial fans with different blade stacking, *AIAA J.* **56**(4) (2018) 1507–1518.
28. R. Ewert, *A Hybrid Computational Aeroacoustics Method to Simulate Airframe Noise* (Shaker Verlag, 2003).
29. P. Martínez-Lera, C. Schram, H. Bériot and R. Hallez, An approach to aerodynamic sound prediction based on incompressible-flow pressure, *J. Sound Vib.* **333**(1) (2014) 132–143.
30. P. J. Roache, Quantification of uncertainty in computational fluid dynamics, *Annu. Rev. Fluid Mech.* **29**(1) (1997) 123–160.
31. I. B. Celik, U. Ghia, P. J. Roache, C. J. Freitas, H. Coleman, P. E. Raad, I. Celik, C. Freitas and H. P. Coleman, Procedure for estimation and reporting of uncertainty due to discretization in CFD applications, *J. Fluids Eng.* (2008).